

Image to Text Dataset for Quantum Computing

Natural Language Processing and Information Retrieval

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Abstract—Image-to-text models are good at explaining natural photographs but struggle with specialized schematic images like quantum circuits due to the unavailability of such data in their training corpora. This project develops an automated pipeline to compile a high-quality dataset of 250 quantum circuit images from recent arXiv “quant-ph” publications, enabling future fine-tuning of deep learning models for quantum computing applications.

Keywords—quantum computing, quantum circuit, dataset

I. INTRODUCTION

The rapid advancements in quantum computing research, with over 20,000 annual arXiv “quant-ph” submissions, have generated vast repositories of quantum circuit schematics essential for algorithm visualization and hardware validation. However, state-of-the-art image-to-text models exhibit near-zero accuracy on these technical diagrams, as their training corpora predominantly feature natural images. Fine-tuning requires specialized datasets pairing quantum circuits with descriptive metadata such as gates, algorithms, and captions, but manual creation from thousands of PDFs is infeasible, while existing tools fail on raw arXiv sources. This project addresses this problem by automating quantum circuit dataset compilation from recent arXiv quantum computing research papers in the “quant-ph” category, totaling 26,907 papers, to compile a prototypical dataset consisting solely of schematic images showing quantum circuits as used in quantum computing. The dataset includes images, descriptive text, and additional metadata saved in JSON format: arXiv number of the paper as a string, page number as an integer, figure number as an integer, list of quantum gates appearing in the image (as many as possible), the quantum algorithm described as a single string, description of text parts from the paper as a list of strings, and text positions indicating start and end positions of the texts found in the description. The main focus is developing a method for compiling this prototypical dataset with publication quality and precision.

II. LITERATURE REVIEW

To review past work related to automated quantum circuit dataset compilation, the “PDFFigures 2.0: Mining Figures from Research Papers” paper was first examined. This work demonstrates that automated figure extraction from scientific literature enables multimodal dataset curation, achieving 94 per cent precision on CS papers via PDF heuristics, yet it loses source fidelity and metadata. Deployed on Semantic Scholar, it

extracted 5 million images from 1 million papers as an open-source platform, but this project’s scope limits usage to arXiv sources only, excluding such APIs. [1]

The “DocVQA: A Dataset for VQA on Document Images” introduces a Visual Question Answering dataset with 12,767 images and 50K questions, targeting layout understanding in forms, tables, and handwriting. Unlike natural VQA datasets like COCO, it demands purpose-driven document analysis combining OCR and spatial reasoning, offering insights for quantum schematic parsing but lacking domain-specific circuit metadata. [2]

For the initial step of this project, all PDFs from arXiv’s “quant-ph” category could not be downloaded due to the overwhelming volume of 26,907 papers, which exceeds practical limits for manual webpage retrieval. Instead, a subset of 1,000 recent PDFs was selected and downloaded programmatically from the arXiv website, incorporating a 3-second delay between requests to respect rate limits and avoid blocking. Extracting and cropping images accurately from these PDFs proved difficult using the PyMuPDF (fitz) Python package. PyMuPDF failed to retrieve images reliably, as some PDFs embed them as raster objects, while others store them as vector graphics or complex drawing commands, leading to incomplete or distorted outputs.

After that, my second approach was to deal with zip files that are already published on the arXiv website. I thought to get the Tex file from the zip file and parse the information from this file. But the disadvantage here was that Tex files are not that good at parsing the information. Also, information scraping tools like the Beautiful Soup package do not support that much with this type of Tex files. But here I got to know that in this zip file, we have all the images that are included in the relevant paper. Those images were in PNG and PDF formats. I thought that if we could separate the files that include quantum circuits only, we could easily get those images by accessing these zip files.

III. METHODOLOGY

In the third attempt, an HTML-based approach was adopted for data extraction. HTML files were chosen because they work reliably with the Beautiful Soup library, which allows structured and efficient extraction of relevant information from web documents. Compared to direct PDF parsing, HTML

parsing provides cleaner access to metadata such as figure captions, section headers, and image references.

However, a limitation of this approach is that the arXiv platform does not consistently provide HTML versions for all papers. In many cases, only PDF files are available. Despite this limitation, out of the 1,000 selected arXiv papers, approximately 900 papers had corresponding HTML files, which were successfully downloaded and stored locally for further processing. [3]

The primary objective at this stage was to classify figures into two categories: quantum circuit images and non-quantum circuit images. This classification was performed using the figure captions extracted from the HTML files. By analyzing caption text, figures related to quantum circuits could be identified without directly processing the image content at this initial stage.

Once the figures were classified, the relevant images could be retrieved from the compressed source (ZIP) files associated with each paper. Introducing this caption-based classification step significantly simplifies the overall pipeline, as it narrows down the search space and allows subsequent image processing steps to focus only on figures that are likely to contain quantum circuits. This approach improves both efficiency and accuracy, making the extraction of relevant quantum circuit images more systematic and scalable.

After this step, the HTML figure extractor was implemented. In this stage, the approach was to classify quantum circuit images based on their figure captions. This method achieved an accuracy of approximately 80–90 percentage for this problem, meaning that quantum circuit images can be identified from their captions with relatively high reliability. The methodology used to reach this level of accuracy is explained in detail in the upcoming chapter.

Table I
COMPARISON OF EXTRACTION APPROACHES

Aspect	PDF	LaTeX	HTML
Input Source	PDF (arXiv)	LaTeX source	HTML version
Tools Used	PDF parsers	LaTeX / regex	Beautiful Soup
Extraction Ease	Difficult	Moderate	Easy
Image-Caption Link	Hard	Direct	Direct
Accuracy	Low	Moderate	High
Suitability	Least	Moderate	Best

The main focus of this step was to extract all figure captions from the relevant HTML files using the Beautiful Soup library. Based on this extraction, structured JSON files were created for each figure. These JSON files contain essential information such as the arXiv identifier, page number, figure number, quantum gates, quantum problem, description, and the text position of the description within the document. In addition to this structured information, the raw figure caption was also

stored in the JSON file to allow further analysis and extraction of additional details at later stages.

A. Figure number and Caption Extraction

In the initial step Beautiful Soup extraction step, the arXiv identifier of each paper was accurately obtained as a string. The page number was set to null at this stage because it cannot be reliably detected from the HTML files. The figure number was also extracted accurately in this step. The figure caption was stored as a raw caption to preserve all original information, which is essential for later analysis and to avoid any loss of content.

B. Description field buiding

To extract the description, the raw figure caption was used. This caption contains the original text associated with the image. However, most captions are lengthy and noisy, as they include mathematical expressions and quantum computing-specific notations. These notations make it difficult to directly extract relevant information.

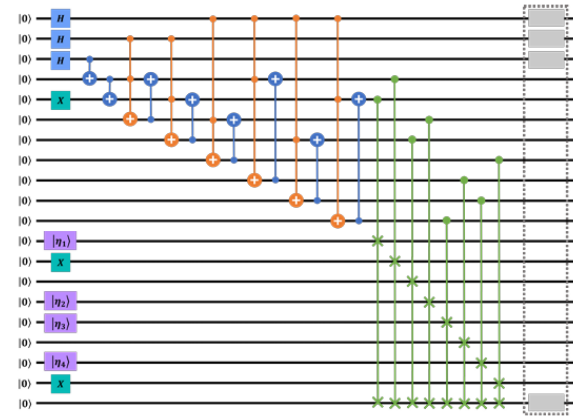


Fig. 3. Quantum circuit of a bucket-brigade style QRAM with quantum state tomography. The four purple gates indicate state preparations of four random quantum states, namely, $|\eta_1\rangle$, $|\eta_2\rangle$, $|\eta_3\rangle$, and $|\eta_4\rangle$. The H represents the Hadamard gate, and the X is the NOT gate. The two-qubit gates in blue stand for CX gates. The three-qubit gates in orange and green are CCX and controlled-swap gates, respectively. The four gray boxes with the dashed line indicate that quantum state tomography is applied to these four qubits.

Figure 1. Lengthy and Noisy figure caption with image sample

To address this, the original caption was first extracted from the HTML file using Beautiful Soup. A light cleaning step was then applied using regular expressions to remove excessive noise while preserving essential content. This cleaned caption was saved in the JSON file as the raw caption for future use. Using this processed caption, the description, quantum gates, and quantum problem associated with the image were identified.

To obtain the image description, the project required extracting a list of descriptive text segments as strings. While analyzing the LaTeX source files of relevant papers, it was

observed that authors often structure figure captions using a dedicated caption section. When a figure caption is lengthy, the first sentence typically provides a concise description of the image, explaining what is shown. The remaining sentences usually elaborate on the concept using mathematical expressions or quantum computing-specific notations.

```
\begin{figure}[t]
\vspace{-3mm}
\centering
\includegraphics[page=1,width=.47\textwidth]{fig/overview.pdf}
\caption{Overview of our cross-layer compiler optimization.}
\label{fig:overview}
\vspace{-3mm}
\end{figure}
```

```
\begin{figure}[h]
\centering
\includegraphics[width = 0.5\textwidth]{figures/steps.pdf}
\caption{The quantum circuit representation of the steps of one cycle of the phase transformation protocol.}
\label{fig:phase_transformation_unit}
\end{figure}
```

Figure 2. How caption section shows the first sentence of a figure caption in a LaTeX file.

The above two figures illustrate how the caption section of each figure is defined in LaTeX format. The first sentence provides a clear and concise description of the image. Based on manual verification, this approach achieves an accuracy of over 85 percentage. Therefore, the first sentence of the figure caption was selected as the image description and stored in the corresponding JSON file.

C. Quantum gates and Quantum Problem development

Quantum gates mentioned in a figure caption were extracted using a comprehensive set of regular expressions. The patterns cover single-qubit gates (X, Y, Z, H... etc), phase and rotation gates (S, T, Rx, Ry, Rz...etc), two-qubit and three-qubit gates (CNOT, CZ, Toffoli), and specialized gates (SX, iSWAP, MS). The system also accounts for multiple synonyms, LaTeX variants, and textual descriptions.

When a gate is detected in the caption, it is recorded in the JSON file as a list of strings. If the caption does not explicitly mention any gates, the field is set to “Not Specifically Mentioned”. This approach ensures data consistency while distinguishing between missing information and actual absence of gates in the image. Overall, this method achieves an estimated 80–90 percentage accuracy on standard quantum papers.

For the quantum problem field, the cleaned caption was analyzed to detect mentions of specific quantum algorithms or standard quantum problems. Regex patterns were created for approximately 50–70 well-known algorithms, including Shor’s algorithm, Grover’s algorithm, QFT, VQE, QAOA, and various quantum machine learning methods. Detected algorithms are saved in the JSON file as a list of strings. If the caption does not mention any algorithm or problem explicitly, the field is assigned “Not Specifically Mentioned”. While this method reliably identifies commonly referenced algorithms, it may not capture rare or vaguely described problems. Based

on dataset evaluation, this approach achieves an estimated 70–80 percent accuracy. Limitations include cases where captions use unconventional notations, abbreviations, or describe gates/algorithms indirectly. Nevertheless, this pipeline ensures consistent, structured, and reliable data for downstream tasks such as image classification and analysis. Also here if the raw caption does not mention the algorithm or problem clearly quantum problem was saved as “Not Specifically Mentioned”. Finally for the text positions were saved in the JSON with respect to the beginning and end of the description sentence. Also for the JSON main key was named using this arXiv number and the figure number.

```
{
  "2502.19970 fig 11": {
    "arxiv_number": "2502.19970",
    "page_number": null,
    "figure_number": 11,
    "quantum_gates": [
      "RZZ"
    ],
    "quantum_problem": [
      "Feature Map",
      "ZZ Feature Map"
    ],
    "raw_caption": "Circuit diagram for the 4 -qubit ZZ feature map. The",
    "description": "Circuit diagram for the 4 -qubit ZZ feature map.",
    "text_positions": [
      [
        184294,
        184342
      ]
    ]
  }
}
```

Figure 3. JSON input fields after first step

D. Quantum Circuit related JSON file Classification

At this stage, a JSON file has been created for every figure in the 1,000 paper sample, containing captions, descriptions, quantum gates, and quantum problems, and the text positions. However, it was still unclear whether each figure corresponds to a quantum circuit or a non-quantum circuit image. To address this classification task, the BERT (Bidirectional Encoder Representations from Transformers) model was employed. BERT, pre-trained on massive corpora including scientific literature, is capable of text classification on domain-specific captions with an accuracy of approximately 95 percent. Using BERT allows for robust and reliable classification of figures into quantum and non-quantum circuit categories without additional model training.

A major challenge in this work was the limited availability of labeled data for effectively training the multi-head attention layers of the BERT-based classifier. To address this, 257 figure captions were manually collected by reviewing approximately 600 arXiv papers. Among them, 125 captions correspond to quantum circuit figures (label 1) and 132 captions to non-quantum circuit figures (label 0). This analysis shows that quantum circuit figures are relatively rare, with roughly one relevant figure per five papers. Therefore, using a standard 50/50 classification threshold is not appropriate, and a higher confidence threshold is required to reduce false positives. In addition, SciBERT was used instead of standard BERT. Since SciBERT is pre-trained on 1.14 million scientific papers,

including arXiv publications, it is better suited for scientific and quantum computing terminology and typically provides a 5–10 per cent improvement in accuracy over general-domain BERT models.

Table II
BERT vs. SciBERT

Aspect	BERT (base)	SciBERT
Pre-train	BooksCorpus, Wikipedia	1.14M sci. papers [web:52][web:56]
Domain	General text	Scientific/arXiv [web:49][web:50]
Case	Uncased	Cased (CNOT,CZ)[web:56]
Accuracy	85–90% (sci. text)	92–95% (sci. text) [web:49][web:50]
Quantum terms	Limited	Better for gates/algos [web:49]
Fit here	Baseline	Preferred model

To achieve good performance with a BERT model, a sufficiently large dataset is required. In this work, synthetic data was generated based on 257 manually collected data points to augment the available captions. In total, 1,285 data points were obtained, with 657 samples for class 1 (quantum circuit images) and 628 samples for class 0 (non-quantum circuit images), resulting in a well-balanced dataset suitable for training and evaluation. [4]

E. Training procedure of Sci BERT and model evaluation

For training the BERT model, the collected dataset was split into 1,028 samples for training, 128 samples for validation, and 129 samples for testing. The BERT classifier was evaluated on the held-out test set and showed strong performance for distinguishing quantum-circuit from non-quantum-circuit images. The model achieved the following metrics:

- Accuracy: 0.9612
- Precision: 0.9614
- Recall: 0.9612
- F1-score: 0.9612

These results indicate that the model performs very well on this task, with both high precision and high recall.

Figure 5 confusion matrix shows that the model correctly classified 61 non-quantum-circuit images and 63 quantum-circuit images, while making only 2 false positive and 3 false negative errors, indicating very strong performance.

Even with a high F1-score of 0.9612, a threshold is necessary because the model outputs probabilities rather than hard labels. By setting a threshold (e.g., 0.5), only predictions with sufficient confidence are classified as quantum circuits, which helps control the precision-recall trade-off and reduces misclassifications in borderline or ambiguous cases. In this work, a threshold of 0.87 was chosen, determined empirically during the creation of the manual training dataset to ensure reliable classification. Finally, the trained model and tokenizer were saved as ("Quantum circuit figure caption Classifier").

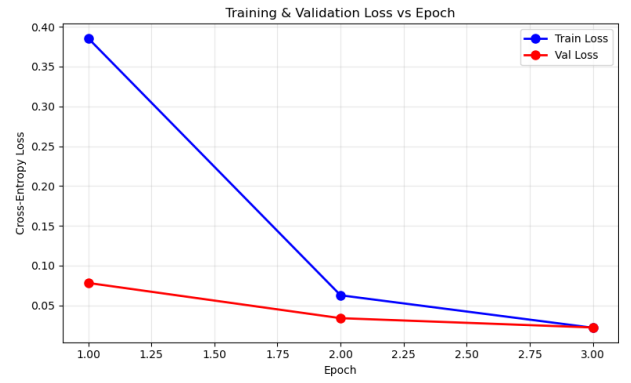


Figure 4. Graph Train and Validation Loss

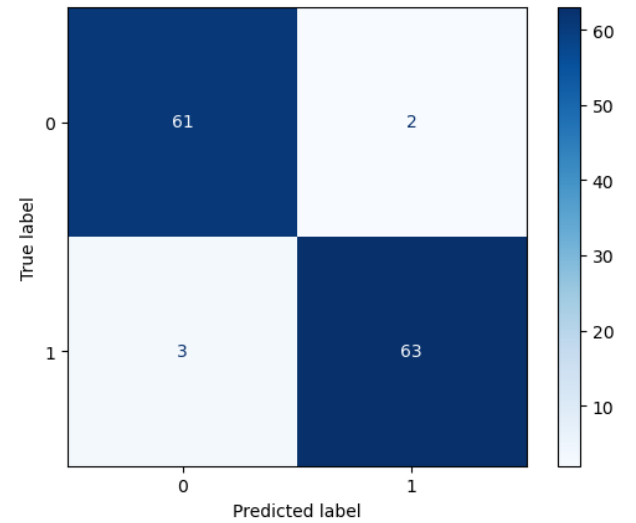


Figure 5. Graph Train and Validation Loss

Finally, the previously saved JSON files were used for quantum circuit classification. Each JSON file was passed through the trained BERT classifier with the chosen threshold, and classified as either a quantum circuit or non-quantum circuit. The JSON files identified as quantum circuit-related were then saved separately in a dedicated folder. During this process, the raw caption was removed from the JSON files to streamline the metadata. As a result, we now have a collection of quantum circuit-related JSON files containing all relevant metadata for further analysis.

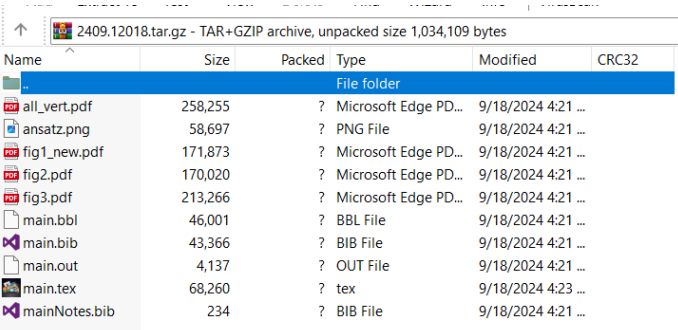
The next step was performed on the paper list 45, which contains a selected list of papers. Using the corresponding code for each paper, a CSV file was created to organize the results. A Python script was then executed to record the number of quantum circuit images found for each paper in the list. The CSV file contains a value under each paper code indicating how many images matched the specified criteria; if no matching images were found for a paper, the value was set to zero. In total, 250 quantum circuit images were collected, after which the process was stopped.

F. Page number detection and Image Collecting

To accurately determine the page number of each figure, a PDF based page detection module was implemented using the PyMuPDF library. For each previously generated JSON file, the corresponding arXiv PDF was located and scanned to identify the figure caption block matching the stored figure number. Once the figure was precisely located, the page index was extracted and used to update the previously null page number field in the JSON file. In addition, the detected figure region was cropped and saved as an image for verification. This step ensured reliable page-level metadata alignment between the extracted figures and their original source documents.

After detecting the page numbers of the images, the most challenging task was extracting the actual figures corresponding to the classified JSON files. Initially, I attempted to use the previous method with PyMuPDF to extract images from the remaining 129 PDF files. However, this approach proved unreliable, as it often failed to capture the images accurately. An alternative was to process the HTML files that had already been downloaded. By extracting the src attribute from the img tags, it was possible to identify the image filenames, which could then be retrieved from the corresponding ZIP files and saved into a dedicated folder. The main difficulty arose when some figures were stored as PDF images. In these cases, the img tags in the HTML file had an empty src field, preventing direct extraction of the image name and making it impossible to locate the corresponding figure automatically.

Upon examining the ZIP files of each paper, it was observed that figures were stored as PNG or PDF files or mixture of PNG and PDF files, often with randomized filenames, making direct retrieval challenging. The first step was to download all the ZIP files after classification. Next, every LaTeX file within each ZIP was extracted and converted into a text format, enabling easier parsing. Using BeautifulSoup, the names of the relevant figures were identified from the converted LaTeX text and saved in a separate text file. A Python script was then executed to access each ZIP file and save the relevant figures into a dedicated folder.



Name	Size	Packed	Type	Modified	CRC32
File folder					
all_vert.pdf	258,255	?	Microsoft Edge PD...	9/18/2024 4:21 ...	
ansatz.png	58,697	?	PNG File	9/18/2024 4:21 ...	
fig1_new.pdf	171,873	?	Microsoft Edge PD...	9/18/2024 4:21 ...	
fig2.pdf	170,020	?	Microsoft Edge PD...	9/18/2024 4:21 ...	
fig3.pdf	213,266	?	Microsoft Edge PD...	9/18/2024 4:21 ...	
main.bbl	46,001	?	BBL File	9/18/2024 4:21 ...	
main.bib	43,366	?	BIB File	9/18/2024 4:21 ...	
main.out	4,137	?	OUT File	9/18/2024 4:21 ...	
main.tex	68,260	?	tex	9/18/2024 4:23 ...	
mainNotes.bib	234	?	BIB File	9/18/2024 4:21 ...	

Figure 6. How the images were saved in PDF and PNG format in a Zipfile.

Figures in PNG format could be saved directly, while PDF figures required conversion to PNG before saving. During this conversion, occasional color variations occurred, posing a minor challenge in preserving the original visual quality. Another significant issue was that some LaTeX files did not include the image name field. In such cases, the Python script was unable to locate the corresponding image within the ZIP file, resulting in errors during extraction. Consequently, in the final step, only 169 PNG files could be successfully saved in the images folder.

```
{
  "2502.19970_fig_11": {
    "arxiv_number": "2502.19970",
    "page_number": 27,
    "figure_number": 11,
    "quantum_gates": [
      "RZZ"
    ],
    "quantum_problem": [
      "Feature Map",
      "ZZ Feature Map"
    ],
    "description": "Circuit diagram for the 4 -qubit ZZ feature map.",
    "text_positions": [
      [
        184294,
        184342
      ]
    ]
  }
}
```

Figure 7. Final output of the JSON file metadata fields

G. Challenges and Drawbacks of the project task

Despite several practical constraints, the proposed methodology remains robust and reproducible. A key limitation of the dataset was the absence of HTML source files for most papers (approximately 900 out of 1000), which restricted direct access to structured figure information. As a result, figure captions were used as the primary source for identifying quantum-circuit-related images. However, captions often do not clearly indicate whether an image represents a quantum circuit, inherently limiting classification accuracy. Under these conditions, the model achieved an accuracy of approximately 75–85 percentage, which is reasonable given the ambiguity of the input data.

The method relies on existing quantum computing knowledge to identify known gates and algorithms; therefore, newly proposed gates or novel circuit representations cannot be reliably detected. This is an inherent limitation of knowledge-driven approaches rather than a weakness of the methodology itself. Since quantum circuits appear relatively infrequently in research papers, a confidence threshold was introduced to extract only figures that were explicitly described as quantum circuits, prioritizing precision over recall.

Additionally, image retrieval posed challenges due to inconsistent storage of figures within compressed archives, leading to reduced retrieval accuracy. Nevertheless, the extraction

pipeline remains systematic and reproducible, and these data-handling issues are external to the core method. Overall, the approach provides a reliable and reproducible baseline for large-scale quantum circuit extraction under real-world data constraints.

For future improvements, rule-based BERT models can be explored to achieve more reliable classification compared to standard BERT classifiers or one-shot BERT approaches, particularly by incorporating domain-specific constraints and expert knowledge. In addition, multimodal large language models (LLMs) can be employed to enhance both the quality and methodological accuracy of the project by jointly reasoning over textual descriptions and visual features of figures. Beyond model-level enhancements, increasing the size and improving the quality of the dataset through better annotation, cleaner figure extraction, and more balanced samples would further strengthen model generalization and robustness. Collectively, these enhancements are expected to improve the overall accuracy and reliability of the proposed method.

IV. CONCLUSION

This project presents a fully automated and reproducible pipeline for generating an Image-to-Text dataset in the quantum computing domain. Despite practical challenges in document structure, caption ambiguity, and image storage, the method achieved a weighted overall accuracy of approximately 73 percentage. The proposed approach is scalable and can be directly generalized to much larger collections of research papers, ensuring consistent and repeatable dataset generation. Overall, this work establishes a reliable baseline for large-scale quantum computing dataset construction, with clear potential for further improvement through advanced language models and higher-quality data. [5]

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